

Nuclear Power Plant CPS Dataset — Full Senior ML Mentor Analysis

Dataset inspected directly: 1,200 rows × 13 columns, 1-minute cadence from 2024-10-16 10:00 to 2024-10-17 05:59, no missing values, no duplicate rows.

1. Dataset Overview

What it is: This is a simulated (near-certainly synthetic) time series of sensor telemetry from a Nuclear Power Plant Cyber-Physical System (CPS) — the kind of dataset used to prototype anomaly/intrusion detection for industrial control systems (ICS) in critical infrastructure. Columns:

Feature	Role
Timestamp	1-minute-resolution index, fully unique, continuous 20-hour window
Reactor Temp (°C)	Core thermal state
Coolant Flow Rate (L/s)	Primary cooling loop throughput
Pressure (MPa)	Primary/secondary circuit pressure
Radiation Level (µSv/h)	Safety-critical exposure reading
Turbine Speed (RPM)	Mechanical/electrical generation state
Pump Status	Categorical (ON/OFF) — 982 ON / 218 OFF
Power Output (MW)	Plant electrical output
Control Rod Position (%)	Reactivity control
Steam Flow Rate (kg/s)	Secondary loop throughput
Vibration Level (mm/s)	Mechanical health indicator
Water Level (m)	Coolant/steam generator level
Anomaly Detected (target)	Binary label, 1,138 "No" / 62 "Yes" (5.17% positive)

Prediction target: Anomaly Detected (binary classification / anomaly detection).

Best-fit ML framing: This is fundamentally a **binary anomaly/rare-event detection problem on multivariate sensor telemetry**, sitting at the intersection of:

- **Classification** (if you trust the labels and just want a detector), and
- **Time-series / unsupervised anomaly detection** (more appropriate given what the data actually shows — see §2).

It is **not** naturally a regression, clustering, recommendation, NLP, or CV problem, though Power Output or Reactor Temp could serve as regression targets for a secondary "digital twin" forecasting task.

2. Data Quality Assessment

Verified directly on the file:

- **Missing values:** None (0 across all 13 columns).
- **Duplicate records:** None (0 exact duplicate rows).
- **Data types:** Clean — all sensor columns are float64, `Pump Status` and `Anomaly Detected` are categorical strings, `Timestamp` is a parseable datetime string. No coercion errors.
- **Class imbalance:** Significant — 94.8% "No" vs 5.2% "Yes" (62 positives out of 1,200). This alone rules out plain accuracy as an evaluation metric and requires resampling (SMOTE) or class-weighting.
- **Outliers:** Sensor ranges look physically plausible on their face (e.g., reactor temp ~300-408°C, pressure ~15-18 MPa) but should still be checked with IQR/Z-score per feature — a few extreme tails exist (e.g., max turbine speed 3822 RPM vs. 75th percentile 3327 RPM).
- **Noise:** Values look like they were generated from smooth distributions with added Gaussian jitter rather than real plant physics (no realistic transients, spikes, or sensor drift patterns typical of real ICS data).
- **Inconsistent values:** None detected structurally — `Pump Status` and `Anomaly Detected` each only take the two expected values.
-  **Critical finding — likely label/feature independence:** Correlation of every numeric sensor feature with the binary target is essentially zero (all $|r| < 0.06$ — e.g., Pressure -0.055, Power Output 0.042, Steam Flow 0.034, Control Rod -0.026, Reactor Temp -0.025, Radiation -0.020, Coolant Flow 0.009, Water Level -0.006, Vibration -0.001, Turbine Speed 0.0007). `Pump Status` also shows no meaningful association with anomalies (Yes-rate 6.4% when ON vs 6.9% when OFF — statistically indistinguishable at this sample size).

- **Implication:** The `Anomaly Detected` label appears to be **randomly/independently assigned** rather than derived from the sensor readings. This is the single most important thing to know before modeling: **any supervised classifier trained on these features will top out around chance-level discrimination (~0.50 ROC-AUC)**, and that result is not a bug — it's the honest empirical finding.
 - **Potential data leakage:** Low risk of classic leakage (no target-derived columns, no post-outcome features), but there IS a subtler risk: because the label doesn't depend on the features, any apparent "signal" a model finds during training is almost certainly **overfitting to noise**, and will not generalize. Watch for this masquerading as a "good" train score with a poor validation score.
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3. Exploratory Data Analysis (EDA)

Important statistics to report:

- Full descriptive stats (mean/std/min/25/50/75/max) per sensor, already computed.
- Class balance table and ratio (18.4:1 negative:positive).
- Point-biserial correlation of each feature vs. target (computed above — all negligible).
- Pump ON/OFF anomaly-rate comparison (6.4% vs 6.9%).

Useful visualizations:

1. Time-series line plots of each sensor over the 20-hour window, with anomaly points overlaid as red markers — visually confirms whether "Yes" rows cluster near any sensor excursions (expect: they won't).
2. Histograms/KDEs of each feature split by class (Yes vs No) — expect near-identical overlapping distributions.
3. Correlation heatmap of all numeric features (including target) — will show the near-zero row/column for the target, visually proving the independence finding.
4. Boxplots per feature by class to check for any residual separation.
5. Pairplot / scatter matrix of the 3-4 most "physically coupled" features (Reactor Temp vs Power Output, Steam Flow vs Turbine Speed) to check internal sensor consistency, independent of the label.
6. Anomaly occurrence timeline (event plot) to check for temporal clustering (bursty anomalies vs. uniformly scattered) — informative even if magnitude features aren't predictive.

Correlation analysis: Feature-feature correlations (e.g., Steam Flow vs Turbine Speed, Reactor Temp vs Power Output) are worth checking to confirm the sensors are physically coherent with each other even though none of them correlate with the label — this

distinguishes "the simulator built a coherent physical system with an independently-sampled anomaly flag" from "the simulator is pure noise everywhere."

Distribution analysis: Check skewness/kurtosis per sensor; most look roughly Gaussian based on quartile spacing.

Hidden patterns: Worth explicitly testing:

- Is the anomaly rate constant over time, or does it change by time-of-day/hour? (chi-square/point process test)
 - Do anomalies cluster in short bursts (autocorrelation of the binary label series) or are they i.i.d. Bernoulli draws? A run-length test will likely show i.i.d. behavior, reinforcing the synthetic-label hypothesis.
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4. Data Preprocessing

- **Cleaning:** Minimal needed — no missing values or duplicates. Still: cast `Timestamp` to datetime, validate physical ranges (e.g., pressure > 0, temp within reactor-safe bounds), and do a formal IQR/Z-score outlier pass per sensor (flag, don't necessarily drop — outliers may matter for unsupervised methods).
- **Encoding:** `Pump Status` → binary (ON=1/OFF=0). `Anomaly Detected` → binary target (Yes=1/No=0).
- **Scaling/Normalization:** StandardScaler for Logistic Regression, One-Class SVM, and LOF (all distance/gradient-sensitive). Tree-based models (Random Forest, XGBoost, Isolation Forest) don't strictly require scaling but scaling doesn't hurt and eases comparison.
- **Feature engineering (worth trying, even knowing the base signal is weak):**
 - Rolling statistics (5/10/15-min rolling mean, std, min, max) per sensor — anomaly detection often depends on *rate of change* rather than absolute value.
 - First-difference / rate-of-change features (Δ Reactor Temp, Δ Pressure per minute).
 - Cross-sensor engineered ratios (e.g., Power Output / Steam Flow, a proxy for turbine efficiency) that might expose inconsistency even if raw sensors don't.
 - Time-of-day / hour-of-day cyclical encoding, in case anomaly rate has time structure.
 - Rolling z-score per sensor (local anomaly deviation), a lightweight per-feature anomaly score useful as an input to a meta-model.
- **Feature selection:** Given the near-zero correlations, standard filter methods (mutual information, ANOVA F-test) should be run and reported transparently — expect all features to score low. Still valuable to include in the paper as evidence.

- **Dimensionality reduction:** PCA/t-SNE/UMAP projections are worth generating specifically to visually show that "Yes" and "No" points are not separable in feature space — a strong visual complement to the correlation table.
 - **Class imbalance handling:** SMOTE on the **training split only** (never on val/test, to avoid leakage of synthetic points into evaluation) — consistent with your supervisor's requirement. Also worth comparing against class-weighted loss and simple random oversampling as ablations.
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5. Machine Learning Opportunities

Best problem formulation: Given the findings, the most defensible framing is a **dual-track study**:

1. **Supervised binary classification** (Logistic Regression, Random Forest, XGBoost) — to empirically demonstrate the label is not learnable from these features.
2. **Unsupervised/semi-supervised anomaly detection** (Isolation Forest, LOF, One-Class SVM) trained only on "normal" sensor behavior, then evaluated against the labels — a fairer test of whether *sensor-defined* anomalies align with *labeled* anomalies.

Models and rationale:

- **Logistic Regression** — interpretable linear baseline; coefficients directly show (near-zero) feature effect sizes.
- **Random Forest** — captures nonlinear interactions if any exist; feature importances double as a second confirmation of the independence finding.
- **XGBoost** — strongest tabular learner available; if even a well-tuned gradient-boosted tree can't beat ~0.50 AUC, that's strong evidence of no learnable signal.
- **Isolation Forest** — unsupervised, isolates points needing fewer splits; well-suited to sensor data without trusting the label.
- **Local Outlier Factor (LOF)** — density-based, catches local (contextual) anomalies that global methods miss.
- **One-Class SVM** — boundary-based novelty detection, useful comparison point to Isolation Forest/LOF.
- **Deep learning possibilities:** An LSTM or Transformer-based autoencoder (reconstruction-error anomaly detection) is a natural extension given the time-series structure — likely to perform on par with classical unsupervised methods here given the small size (1,200 rows) and lack of real learnable signal, but it's a legitimate "we

tried it" section for a paper, especially if you frame it as testing whether temporal context recovers signal that point-wise models miss.

- **Ensemble methods:** A stacked/voting ensemble of the three unsupervised detectors (Isolation Forest + LOF + One-Class SVM anomaly scores) is a strong "practical system" contribution, even if none individually beats chance against the given labels — it's a defensible real-world design pattern.
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6. Evaluation Metrics

Given severe class imbalance (5.2% positive) and the anomaly-detection framing:

- **ROC-AUC** — primary metric, threshold-independent, already your reported headline number (~ 0.50 = chance).
 - **PR-AUC (Average Precision)** — arguably *more* informative than ROC-AUC here because of the imbalance; report it alongside ROC-AUC since ROC-AUC can look deceptively reasonable under heavy imbalance while PR-AUC exposes poor performance more starkly.
 - **Precision, Recall, F1 (positive class)** — essential for a rare-event detector; recall matters most in safety-critical nuclear contexts (missed anomalies are costlier than false alarms), so report a recall-oriented operating point too.
 - **Confusion matrix at a chosen threshold** — always include, since a single scalar metric hides the 62-vs-1138 imbalance reality.
 - **Balanced Accuracy** — useful supplementary metric since raw accuracy is meaningless here (predicting all-"No" gives 94.8% accuracy).
 - **MCC (Matthews Correlation Coefficient)** — a single balanced summary stat that's robust to imbalance; recommended in addition to F1.
 - Accuracy alone should be explicitly called out in your paper as **misleading** for this dataset — this is a good methodological talking point.
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7. Project Ideas (easiest → most advanced)

1. **Baseline EDA + classical classifiers report** (LogReg, RF) with a clear "here's what accuracy hides" narrative.
2. **Class-imbalance handling comparison:** SMOTE vs class-weighting vs random oversampling vs undersampling, benchmarked on the same models.
3. **Unsupervised anomaly detector comparison:** Isolation Forest vs LOF vs One-Class SVM, scored against labels.

4. **Feature importance / interpretability study** (SHAP values on RF/XGBoost) to explain *why* nothing is predictive.
 5. **Synthetic label detection meta-study**: statistically test and document that the labels appear independent of features (permutation tests, mutual information, distribution overlap tests) — this can become the paper's central contribution.
 6. **Rolling-window feature engineering study**: does adding temporal derivatives/rolling stats change the picture at all?
 7. **Dimensionality reduction visualization suite** (PCA/UMAP/t-SNE) as a diagnostic tool for "is this label learnable" — a reusable technique for other tabular datasets.
 8. **Ensemble anomaly-scoring system**: combine multiple unsupervised detectors into one alert score for a more realistic ICS monitoring pipeline.
 9. **Deep learning autoencoder / LSTM reconstruction-error detector** on the multivariate time series.
 10. **Full "trustworthy AI" case study**: a methodology paper on how to detect and report unlearnable/synthetic labels in ML datasets before wasting compute on model tuning — positions your ~0.50 AUC finding as the actual scientific contribution rather than a failure.
 11. **End-to-end deployable ICS monitoring dashboard** combining unsupervised scoring + rule-based physical thresholds + supervised layer, with full MLOps packaging (most advanced, portfolio-capstone level).
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8. Research Opportunities

- **Novel angle**: Reframe the "near-0.50 AUC across 6 models" finding as a **methodological contribution on label reliability auditing** — proposing a standardized pre-modeling diagnostic (correlation screening + permutation test + mutual information + visual separability via UMAP) that researchers should run *before* reporting model benchmarks on any new anomaly-detection dataset.
- **Research gap**: Many published ICS/CPS anomaly-detection papers using synthetic datasets don't verify that labels are causally tied to features; your work provides a counter-example and a checklist to catch this.
- **Improvement over existing approaches**: Most anomaly-detection papers report only the "best" model's score; explicitly reporting *all* six models converging to chance-level, plus the diagnostic evidence *why*, is a stronger and more transparent scientific narrative than typical benchmark papers.
- **Conference-worthy framing**: Position as a **negative-results / reproducibility paper** ("On the Reliability of Synthetic Labels in Cyber-Physical Anomaly Detection Benchmarks") — these are increasingly welcomed at venues focused on ML reliability, dataset auditing, or ICS security (e.g., workshops at NeurIPS/ICML Datasets &

Benchmarks track, IEEE conferences on industrial cyber-physical systems, or security-focused venues like ACM CCS workshops).

- **Follow-up research:** Compare against a real or higher-fidelity nuclear/ICS dataset (e.g., publicly available SCADA/industrial testbed datasets) to show the diagnostic checklist correctly identifies *learnable* labels there, proving your method isn't just detecting "this dataset happens to be bad" but generalizes as a useful audit tool.
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9. Deployment Ideas

- **Web application:** A Streamlit/Gradio dashboard where a user uploads sensor logs and gets live anomaly scores from the unsupervised ensemble, plus a "label reliability" diagnostic report.
 - **API:** A FastAPI microservice wrapping the trained Isolation Forest/LOF ensemble, returning anomaly probability + confidence + feature contribution (SHAP) per request — deployable behind a real ICS data historian.
 - **Mobile app:** A lightweight alert-viewer app for plant operators, pushing notifications when the anomaly score crosses a threshold (would need a lighter-weight model, e.g., a small Isolation Forest, for on-device or edge inference).
 - **Dashboard:** A live Grafana/Plotly Dash monitoring board with per-sensor time series, rolling anomaly score, and historical event log — realistic for a SCADA-adjacent operations center.
 - **Real-world system:** As a component in a broader **defense-in-depth ICS security stack**, combined with rule-based physical safety interlocks (never rely on ML alone for nuclear safety-critical alerts) — worth stating explicitly in any paper/deployment writeup as an ethical/safety caveat.
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10. Improvement Suggestions

Additional data that would help:

- Ground-truth, physically-derived anomaly labels (e.g., generated from actual out-of-spec sensor thresholds or known fault-injection events), rather than what appears to be randomly sampled labels.
- Longer time horizon (weeks/months, not 20 hours) to capture rare real-world fault regimes and seasonal/operational cycles.
- Metadata: fault type/category, severity level, root cause, maintenance logs, operator actions/annotations.

- Higher sampling frequency (sub-minute) for vibration/radiation, which are typically monitored at higher resolution in real plants.
- Network/cyber telemetry (if this is meant to model a cyberattack scenario) — e.g., command/control traffic logs, so "cyber" anomalies (spoofed sensor writes, replay attacks) are distinguishable from "physical" faults.

New features to engineer:

- Rolling statistics and rate-of-change per sensor (see §4).
 - Cross-sensor physical-consistency residuals (e.g., expected Power Output given Steam Flow & Turbine Speed via a physics-informed regression; anomaly = large residual).
 - Time-since-last-anomaly and time-of-day features.
 - Composite "system stress index" combining normalized deviations across all sensors.
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11. Complete ML Pipeline (Roadmap)

1. **Data ingestion & validation** — load CSV, validate dtypes/ranges, check for missing/duplicates (done — clean).
2. **EDA** — full descriptive stats, distribution plots, correlation heatmap, class balance report.
3. **Label reliability audit** — correlation, mutual information, permutation test, PCA/UMAP visual separability (your key methodological step).
4. **Preprocessing** — encode categoricals, scale numerics, engineer rolling/derivative features.
5. **Stratified 70/15/15 train/val/test split** (per supervisor requirement) — apply SMOTE to training fold only.
6. **Baseline supervised modeling** — Logistic Regression, Random Forest, XGBoost with cross-validation on the train/val split.
7. **Unsupervised modeling** — Isolation Forest, LOF, One-Class SVM trained on "normal"-labeled training data.
8. **Model evaluation** — ROC-AUC, PR-AUC, F1, MCC, confusion matrices on held-out test set; report honestly even if near-chance.
9. **Interpretability** — SHAP/feature importance for supervised models; explain the near-zero attributions.
10. **Ensembling** — combine unsupervised anomaly scores into a single alert system.
11. **Reporting/visualization** — 12+ publication-quality figures, result tables (already produced per your prior work).

12. **Packaging** — reproducible Colab-compatible notebook, requirements file, README.
 13. **(Optional) Deployment** — FastAPI/Streamlit demo wrapping the final ensemble.
 14. **Manuscript writing** — frame the null result as the central, transparent scientific contribution.
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12. Code Roadmap (implementation order in Python)

1. `pandas`/`numpy` — load, validate, clean.
 2. `matplotlib`/`seaborn`/`plotly` — EDA visualizations.
 3. `scipy.stats` — correlation tests, permutation tests, mutual information (`sklearn.feature_selection.mutual_info_classif`).
 4. `sklearn.preprocessing` — `StandardScaler`, `OneHotEncoder`/manual binary mapping.
 5. `sklearn.model_selection.train_test_split` (stratified, twice, for 70/15/15).
 6. `imblearn.over_sampling.SMOTE` — training-fold only.
 7. `sklearn.linear_model.LogisticRegression`, `sklearn.ensemble.RandomForestClassifier`, `xgboost.XGBClassifier` — supervised baselines.
 8. `sklearn.ensemble.IsolationForest`, `sklearn.neighbors.LocalOutlierFactor`, `sklearn.svm.OneClassSVM` — unsupervised detectors.
 9. `sklearn.metrics` — `roc_auc_score`, `average_precision_score`, `f1_score`, `matthews_corrcoef`, `confusion_matrix`.
 10. `shap` — interpretability.
 11. `sklearn.decomposition.PCA`, `umap-learn` — dimensionality reduction visuals.
 12. (Optional) `tensorflow`/`torch` — LSTM/Transformer autoencoder for the deep-learning extension.
 13. `fastapi`/`streamlit` — deployment demo (optional/stretch).
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13. Risks and Challenges

- **Over-tuning to noise:** With a label independent of the features, aggressive hyperparameter search *will* eventually find a combination that looks good on validation by chance — always confirm on a truly held-out test set, and be suspicious of any AUC meaningfully above ~0.55-0.60.
- **Misleading accuracy:** A naive "always predict No" model scores 94.8% accuracy — never lead with accuracy in your paper.

- **SMOTE leakage:** Applying SMOTE before splitting (or on val/test) inflates performance artificially — apply strictly after the split, on the training fold only.
 - **Small positive class (n=62):** Limits statistical power for both training and evaluation; be cautious interpreting any subgroup analysis on the positive class, and consider bootstrapped confidence intervals on your reported metrics.
 - **Overclaiming novelty:** Be careful not to frame "we tried 6 models" alone as a contribution — the contribution is the diagnostic methodology and the transparent reporting of a null result, which is scientifically valuable but must be framed correctly to avoid looking like a failed experiment rather than a deliberate finding.
 - **Reviewer skepticism of null results:** Some reviewers instinctively discount "nothing worked" papers — mitigate by strongly emphasizing the generalizable auditing methodology and comparing against a dataset where your method *does* detect learnable signal.
 - **Safety framing:** Since this is nuclear-domain data, explicitly state (in any public-facing report) that this is a synthetic/simulated dataset only, not derived from or applicable to real plant safety systems, to avoid misinterpretation.
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14. Final Recommendation

If I were building the best possible portfolio/publication project from this exact dataset, I would **not** chase a higher AUC number — the data doesn't support one, and pretending otherwise would be scientifically dishonest. Instead I would build:

"A Label-Reliability Auditing Framework for Cyber-Physical Anomaly Detection Benchmarks," structured as:

1. A rigorous, transparent benchmark of 6 classical models (already done) showing convergence to chance-level performance.
2. A **novel diagnostic toolkit** (correlation screen + mutual information + permutation test + UMAP visual separability + unsupervised-detector cross-check) packaged as reusable functions/a small Python utility — this is the differentiated, reusable contribution.
3. A validation of that toolkit against a second, real or higher-fidelity dataset, showing it correctly flags "this one has learnable signal."
4. A polished notebook + paper draft that frames the ~0.50 ROC-AUC finding not as a dead end but as *evidence the toolkit works as intended* — catching a bad dataset before wasted effort.

This is stronger than a typical "trained 6 models, got 0.95 accuracy" student project because it demonstrates statistical maturity, methodological rigor, and exactly the kind of critical

thinking (not taking a benchmark at face value) that supervisors and reviewers respect — and it directly extends the transparent finding your prior notebook already established.